THE CANONICAL MAP BETWEEN FIBRE PRODUCTS AFTER CHANGING THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4464-4476, following the statement at 4454-4462.

Editorial note. The fibre-product identification 'by general category theory' is anchored by the displayed diagram; each conclusion is dispatched to a named engine.

Allowed standard prerequisites (editorial).

- The relative diagonal is an immersion. lemma-diagonal-immersion (card L26408, cited at ~4473). Delta_{T/S} is an immersion (closed if T/S separated).
- Immersions stable under fibre products with a fixed factor. lemma-fibre-product-immersion (card L26413, cited at ~4474). Base change of the diagonal is an immersion.
- Base change preserves quasi-compactness. lemma-quasi-compact-preserved-base-change (same chapter, cited at ~4475). Quasi-separated case.

Source reference. Original lemma 01KJ: sources/stacks-schemes/pinned-original-source.tex, source lines 4255–4255. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01JU: sources/stacks-schemes/pinned-original-source.tex, source lines 3547–3547. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01K5: sources/stacks-schemes/pinned-original-source.tex, source lines 3843–3843. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to T$ and $g : Y \to T$ be morphisms of schemes with the same target. Let $h : T \to S$ be a morphism of schemes. Then the induced morphism $i : X \times_T Y \to X \times_S Y$ is an immersion. If $T \to S$ is separated, then i is a closed immersion. If $T \to S$ is quasi-separated, then i is a quasi-compact morphism.*

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Proof. By general category theory the following diagram

$$\begin{array}{ccc} X \times_T Y & \longrightarrow & X \times_S Y \\ \downarrow & & \downarrow \\ T & \xrightarrow{\Delta_{T/S}} & T \times_S T \end{array}$$

is a fibre product diagram. The lemma follows from Lemmas 01KJ, 01JU and 01K5. □